Rigidity of the dancing conformal structure (Theorem 2.3)

Dunajski, Maciej

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/sigma-2022-027-twistor-theory-of-dancing-paths/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Printed numbers anchored to the published PDF (SIGMA 18 (2022) 027, stream-extracted text sigma22-027.pdf, census pdf_anchor_evidence): Definition 2.1/2.2, Theorem 2.3/2.4/2.5 in section 2, Proposition 3.1 in section 3, Lemma 4.1 in section 4, Remark 6.1 in section 6, confirming the sectionwise shared master counter Theorem declared in the author's newtheorem block (source lines 11-19: Corollary, Lemma, Proposition, Definition, Example, Remark all hang on the Theorem counter).

Editorial note. Theorem-family numbers and equation numbers inside the card are locally renumbered (contexts taken from other sections of the source cannot keep their paper numbers under the sectionwise counters); every cross-reference is a symbolic ref and resolves inside the card.

Editorial note. The counted span is the author's named proof environment 'Proof of Theorem \ref{theo_rigid}' (source lines 449-541), a remote proof declared in section 3 (008 remote-named-proof precedent).

Editorial note. Contexts delivered in source order: double_fib (2.1), definull (2.2), lagrangian (2.2), the affine_c statement, the section-3 opening framework prose, the total-null definition, the alpha_s statement, the beta-surface definition, and the bridge sentence; all locally renumbered.

Editorial note. The proof states (source lines 531-532) that one equality 'follows from a rather complicated computation which we have performed on MAPLE' - the author's own disclosure, kept verbatim.

Editorial note. Context [437,442] ends before the sentence 'The result of Lemma 4.1 fits into the framework of half-flat causal structures' (source lines 443-444), which carries two citations outside this card's cite closure and is not needed by the proof.

Editorial scope. Single environment card cut from the SIGMA article doi:10.3842/SIGMA.2022.027 (sigma-2022-027-twistor-theory-of-dancing-paths); statement and proof are the author's own text at source lines 255-261 and 449-541.

Allowed standard prerequisites (editorial).

- Alpha-surface family (Lemma 4.1). Delivered in context (source lines 413-423).. Load-bearing: the conformal class is anti-self-dual because the alpha-surfaces are null (source lines 450-456).
- Ward theorem with cosmological constant. Cited in the proof (Ward_cosmo).. Load-bearing: contact structure on the twistor space implies an Einstein metric (source lines 461-463).
- Nonlinear graviton theorem. Cited in the proof (Penrose).. Load-bearing: ASD Weyl tensor from the alpha-surface family (source lines 455-456).

Theorem 2.3. *If the dancing condition on a path geometry $\mathcal{U}$ defines a conformal structure on $M = (\mathcal{U}\times\mathcal{Z})\backslash\mathcal{I}$, then $\mathcal{U} = \mathbb{RP}^2$ with its flat projective structure, and the conformal structure is represented by the* SL(3)*-invariant anti-self-dual Einstein metric on $M = \mathrm{SL}(3)/\mathrm{GL}(2)$.*

$$\mathcal{U} \xleftarrow{\mu} \mathcal{I} \xrightarrow{\nu} \mathcal{Z}, \tag{2.1}$$

Definition 2.4. A tangent vector $(\dot{u}, \dot{z}) \in T_{(u,z)}M$ is called *null* if the path through u in the direction of $\dot{u}$ contains the turning point of z.

$$\mathcal{L}(u, z, \dot{u}, \dot{z}) = 0, \tag{2.2}$$

Proposition 2.5. *There exist affine coordinates $\left(x^0, x^1, \zeta_0, \zeta_1\right)$ on $M = \mathrm{SL}(3)/\mathrm{GL}(2)$ such that two points are dancing if and only if they are null separated with respect to the anti-self-dual Einsten metric*

$$g = \mathrm{d}\zeta_0\mathrm{d}x^0 + \mathrm{d}\zeta_1\mathrm{d}x^1 + (\zeta\cdot\mathrm{d}x)^2, \qquad \textit{where} \quad \zeta\cdot\mathrm{d}x \equiv \zeta_0\mathrm{d}x^0 + \zeta_1\mathrm{d}x^1. \tag{2.3}$$

The incidence relation encoding the double fibration (2.1) is given by a map $\psi\colon \mathcal{U}\times\mathcal{Z}\to\mathbb{R}$, where $\mathcal{I}$ is the inverse image $\psi^{-1}(0)\subset\mathcal{U}\times\mathcal{Z}$. Let $(u, z)\in M$, i.e., $\psi(u, z)\neq 0$, and let $(\dot{u}, \dot{z})\in T_{(u,z)}M$. The infinitesimal dancing condition is equivalent to the existence of a pair (u^*, z^*) such that

$$\psi(u, z^*) = 0, \quad \psi(u^*, z) = 0, \quad \psi(u^*, z^*) = 0, \quad \dot{u}\frac{\partial\psi}{\partial u}(u, z^*) = 0, \quad \dot{z}\frac{\partial\psi}{\partial z}(u^*, z) = 0,$$

where z^* is the path in the direction $\dot{u}$ containing u and the turning point u^* of the path z. Here the notation $\dot{u}\partial/\partial u$ denotes a differentiation along a vector field.

We now eliminate z^* between the 1st and the 4th condition, and eliminate u^* between the 2nd and the 5th condition. Substituting the resulting expressions to the 3rd condition gives one relation between $(u, z, \dot{u}, \dot{z})$ which is of the form (2.2).

We shall say that a two-dimensional surface $\sigma\subset M$ is totally null if all tangent vector fields in $T_m\sigma, m\in M$ are null in the sense of Definition 2.4.

Lemma 2.6. *There exists a three-parameter family of totally null surfaces (call them α-surfaces) in M which correspond to points in $\mathcal{I}$. For any $(l, p)\subset\mathcal{I}$ the corresponding α-surface consists of $(L, P)\in M$ such that*

$$P\subset l, \qquad p\subset L, \qquad P\neq p, \qquad L\neq l. \tag{2.4}$$

In addition to the α surfaces M also admits two two-parameter families of null surfaces (call them β-surfaces) defined as follows. The first family β_1 consist of a fixed point $P \in \mathcal{U}$ (hence two parameters), and all lines L in $\mathcal{U}$ not containing P. The second family β_2 consist of a fixed line $L \subset \mathcal{U}$ and all points P not on L. If we have instead chosen to model M on the points and paths in $(\mathcal{Z}, \mathfrak{z})$, then the roles of β_1 and β_2 would swap, but the notion

We are now ready to prove Theorem 2.3, and thus establish the uniqueness of the dancing metric.

Proof of Theorem 2.3. If there exists a conformal structure on M, then, by Lemma 2.6 it is necessarily anti-self-dual (with the right choice of the orientation). Indeed, there exists a three-parameter family of null surfaces. All null surfaces in four dimensions are self-dual (SD) or anti-self-dual (ASD), and we choose an orientation of M such that the α-surfaces are SD. Thus the Weyl tensor of any metric g in the conformal class $[g]$ is ASD by Penrose's nonlinear graviton theorem [5] adapted to neutral signature.

Now we shall argue that the conformal structure $[g]$ defined by the α-surfaces of Lemma 2.6 must contain an Einstein metric. To show it, it is enough to demonstrate that the twistor space $\mathcal{I}$ of incident pairs (α-surfaces) admits a contact structure. The existence of an Einstein metric will then be a consequence of the theorem of Ward [6]. The contact structure is intrinsically defined by the double-fibration picture (2.1). The space $\mathcal{I}$ is equipped with two one-dimensional distributions defining the quotients to $\mathcal{U}$ and $\mathcal{Z}$. The resulting two-dimensional distribution is non-integrable, and thus it defines a one-form θ on $\mathcal{I}$ such that $\theta \wedge \mathrm{d}\theta \neq 0$.

We have shown (after the proof of Lemma 2.6) that on top of the SD α-surfaces there exist the two, two-parameter family of β-surfaces. Let $(\mathbb{S}, \epsilon)$, and $(\mathbb{S}, \epsilon')$ be two symplectic real vector bundles over M (see [1, 2]). We shall make use of the two isomorphisms: $TM = \mathbb{S} \otimes \mathbb{S}'$, and $\Lambda^2{}_- = \mathbb{S} \odot \mathbb{S}$, where $\Lambda^2{}_-$ is the rank-3 vector bundle of ASD two-forms. Let the two families of β surfaces correspond to ASD two-forms Σ_1 and Σ_2 in $\Gamma\big(\Lambda^2{}_-\big)$, or equivalently to two sections ι, $\tilde{\iota}$ of $\mathbb{S}$.

We can rescale ι so that the corresponding two-form is closed, and proportional to $\mathrm{d}x^1 \wedge \mathrm{d}x^2$ for some functions $\big(x^1, x^2\big)$ on M which are constant on each β-surface in the first two parameter family. The Frobenius integrability conditions now imply the local existence of two more coordinates such that ζ_0 and ζ_1 on M such that $\mathrm{Ker}(\Sigma_1) = \mathrm{span}\{\partial/\partial\zeta_0, \partial/\partial\zeta_1\}$. The functions (ζ_0, ζ_1) are then the coordinates on the β-surface. The corresponding metric takes the form

$$g = \mathrm{d}\zeta_A \odot \mathrm{d}x^A + \Theta_{AB}(x, \zeta)\mathrm{d}x^A \odot \mathrm{d}x^B, \qquad A, B = 0, 1 \tag{2.5}$$

for some symmetric two-by-two matrix Θ. The anti-self-duality condition on the Weyl tensor forces the components of Θ to be at most cubic in (ζ_0, ζ_1), with additional algebraic relations between the components. Imposing the Einstein condition gives

$$\Theta_{AB} = \zeta_A\zeta_B + \mathrm{P}_{(AB)} - \Gamma^C_{AB}\zeta_C, \tag{2.6}$$

where the coefficients Γ^C_{AB} only depend on $\big(x^0, x^1\big)$, and P_{AB} is the projective Schouten tensor of affine connection ∇ with connection components Γ^C_{AB}. The β-distribution defined by Σ_1 is parallel, and

$$\nabla^g\Sigma_1 = 6\mathcal{A} \otimes \Sigma_1,$$

where ∇^g is the Levi-Civita connection of g, and the one-form $\mathcal{A}$ is a symplectic connection for the symplectic structure structure [4] given by

$$\Omega = \mathrm{d}\mathcal{A} = \mathrm{d}\zeta_A \wedge \mathrm{d}x^A + \mathrm{P}_{AB}\mathrm{d}x^A \wedge \mathrm{d}x^B. \tag{2.7}$$

Up to this point our computation has followed [4]. Now consider the second family of β surfaces given by the ASD two-form Σ_2, which in our chosen coordinate system is given by

$$\Sigma_2 = \epsilon^{AB}\big(\mathrm{d}\zeta_A + \Theta_{AC}\mathrm{d}x^C\big) \wedge \big(\mathrm{d}\zeta_B + \Theta_{BD}\mathrm{d}x^D\big).$$

The normalisation condition $\epsilon(\iota, \tilde{\iota}) = 1$, and the condition that the distribution defined by Σ_2 is parallel together imply that this two-form satisfies

$$\nabla^g\Sigma_2 = -6\mathcal{A} \otimes \Sigma_2,$$

where $\mathcal{A}$ is given by (2.7). We will need only the skew-symmetric part of this equation, which gives

$$0 = \mathrm{d}\Sigma_2 + 6\mathcal{A} \wedge \Sigma_2 = \epsilon^{CD}\nabla_{[A}\mathrm{P}_{B]C}\mathrm{d}x^A \wedge \mathrm{d}x^B \wedge \mathrm{d}\zeta_D,$$

where the second equality follows from a rather complicated computation which we have performed on MAPLE. Therefore this second family of β-surfaces exists iff $\nabla_{[A}P_{B]C} = 0$, which (for two-dimensional projective structure) is precisely the condition (see, e.g., [3]) that the connection ∇ is projectively equivalent to a flat connection, where we can chose $\Gamma^C_{AB} = 0$ and $\mathrm{P}_{AB} = 0$. It was shown in [4] that the metric (2.5) with Θ_{AB} given by (2.6) is invariant under projective changes of connection complemented by an affine translation of ξ_A. Therefore the resulting metric is of the form (2.3), which completes the proof. □

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